

kta Process Map

Compiled proof of work - vyakaranamlikhitam/kta - 6 August 2026

Purpose

A long-term derivational index for deciding the Sanskrit kta form of a dhatu. Rules and explanations are stored once and reused by stable hyperlinks; examples accumulate by gana without forcing repeated prose or structural rework.

Decision map

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|--------------------|--|---|
| 1. Root identity | | Resolve the exact Dhatupatha entry: gana, indicatory markers, meaning, pada, preverb, and intended interpretation. |
| 2. Affix | | Introduce kta under A. 3.2.102 and register kta/ktavatu as nistha under A. 1.1.26. |
| 3. Interpretation | | Test the ordinary passive/resultative reading, then kartari use for motion roots, intransitives, and listed roots under A. 3.4.72. |
| 4. it | | Apply the ardhadhatuka it system beginning with A. 7.2.35; gana membership alone does not decide set, anit, or vet behavior. |
| 5. Stem operations | | Apply root-specific substitutions and deletions in grammatical order. Every operation receives a local usage anchor. |
| 6. Nistha rules | | Search A. 8.2.42ff for n-substitution, lexical prescription, optionality, prohibition, and meaning-conditioned forms. |
| 7. Sandhi | | Apply only operations actually triggered; each links first to a local use and then to the ordered global rule registry. |
| 8. Audit | | Return the surface form, derivation path, competing licensed forms, interpretation, source status, and unresolved commentary questions. |

Repository architecture

Stable targets survive reordering and regeneration

Preamble 1 - dhatugana

Generated from the complete Dhatupatha. Each root receives an identifier such as DHATU-01-0001.

Preamble 2 - sutrani

Pāṇinian rules are sorted numerically. Each entry records scope, operation, status, and commentary/audit notes.

Chapters 1-11 - ganapadani

Every gana chapter begins with "sutras used in this section". A derivation links to the local usage anchor first, then to the global rule entry.

Cross-cutting registries

Sandhi rules employed; composition categories with dhatu lists; irregular constructions by gana; audit ledger.

Proof convention

Every exploratory project terminates in vyakaranamlikhitam/<project>/ containing only compiled PDF artifacts.

Representative derivation models

Form	Audit status
bhu -> bhuta	intransitive/kartari model
gam -> gata	motion-root kartari model
pac -> pakva	lexical nistha prescription
vid -> vitta / vidita	optionality and meaning must be separated
sush -> shushka	special k-prescription
kr -> kṛta	canonical passive/resultative model
kanduy -> kanduyita	it-bearing model

Archive path

```
vyakaranamlikhitam/  
    kta/  
        kta-process-map-proof.pdf
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